

Simulation

CSU, Chico Math 350

2026-02-02

outline

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Useful R Operators

For logical statements, the following operators are very helpful:

AND: &

OR: |

NOT: !

Examples using the email data set from the openintro package:

```
mean(email$spam & email$format)
# probability an email is spam AND HTML

## [1] 0.04029584

mean(!email$spam)
# probability an email is NOT spam

## [1] 0.9064014

mean(!email$spam | !email$format)
# probability an email is either NOT spam OR NOT HTML

## [1] 0.9597042
```

Defining Vectors in R

Some tips to help with `sample()` problems:

- ▶ The function `c()` (short for concatenate) combines several items into a single object. Separate each item with a comma.
Example: `x <- c(1,5,10)`
- ▶ A vector with words or text strings must have quotes around each one for R to treat it as text.
Example: `y <- c("UnderAvg", "Avg", "OverAvg")`
- ▶ Do not use commas in numbers (ex: 10,000) as R will not read them properly.
- ▶ Variable names cannot begin with a number or symbol (`2x` is not a valid name, but `x2` is). Avoid using certain names (such as `T` or `F`).
- ▶ Keep variable names short but descriptive.
Example: `"equals6"` instead of just `"e"` or `"doesthesumequal6"`.

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Hands on with `sample()`

Now that we know a few basic terms and tools, let's get into some examples for practice.

Sampling from a Vector

Take a sample of two numbers from the numbers 1 through 10 without replacement.

```
x <- 1:10  
sample(x, size = 2)
```

First we define the vector. As a reminder, 1:10 produces all integers between 1 and 10, inclusive. We then sample from the vector, setting the sample size to equal 2.

Example 01: Sampling Months

Now try filling in this one:

Sample without replacement three months from the list of months.

```
months <- c("Jan","Feb","Mar","Apr","May","Jun",  
"Jul","Aug","Sept","Oct","Nov","Dec")  
sample(_____,size = ____)
```

You can directly copy the code here to RStudio. Please fill in the blanks (by replacing the underscores with your code) and report the output of your code on CodiMD.

Sampling with Replacement

The defaults can be changed. For instance, if you want to sample with replacement you would just add `replace=TRUE` to the command. This allows you to sample more values than the size of the sample space.

Below: Sample 10 times from the vector 1:5.

```
x <- 1:5  
sample(x, size = 10, replace = TRUE)
```

Example: Sampling Days

Here's another to try:

Create a sample of 10 random days of the week. Just use the first two letters in the day name, and all lowercase (you will need to define this vector).

```
days <- _____  
sample(_____,size = _____, replace = _____)
```

Once again, please fill in the blanks and report the output of your code on CodiMD.

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Dice Roll Events

Suppose that two six-sided dice are rolled and the numbers appearing on the dice are added. Calculate the probabilities of the two events listed below using both theoretical methods and simulation.

Event D: The sum of the two dice is 6.

Event E: At least 1 die is a 2.

Event D, Part 1

A successful simulation is built up in stages. First you create the sample space, then the event space. Then you identify the proportion of values drawn from the sample space that fall into the event space.

1. Create two vectors, each of length 10,000, by sampling the numbers from 1 to 6. These represent the rolls on each of two dice.

```
die_1 <- sample(x=___, size=___, replace=___)
die_2 <- sample(x=___, size=___, replace=___)
```

Event D, Part 2

2. Add these two vectors of dice results together to create the sample space.

```
die_1 <- sample(x=___, size=___, replace=___)
die_2 <- sample(x=___, size=___, replace=___)
sum.of.2.dice <- ___ + ___
```

Event D, Part 3

3. Create a boolean value (that is, TRUE or FALSE) that represents whether or not the sample space is in the event space, then calculate the mean of the boolean value.

```
die_1 <- sample(x=___, size=___, replace=___)
die_2 <- sample(x=___, size=___, replace=___)
sum.of.2.dice <- ___ + ___
D <- sum.of.2.dice == ___
mean(___)
```


Event D, End

Now fill in the blanks and share your result on CodiMD!
Everyone's answer will be different, but they should all be fairly close to the theoretical value.

Event E

This time, we'll need to use the logic operators from the first slide to determine the probability of this event.

```
die_1 <- ____  
die_2 <- ____  
E <- die_1==____ | die_2 == ____  
mean(E)
```

After filling in the blanks, please list your result on CodiMD.

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Roll three six-sided dice and add all the face up numbers. Use simulation to estimate the probability that the sum of the three dice is at least 12.

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The replicate() Function

We simulated experiments using `sample()` to get random samples of a particular size, but sometimes our expressions are a bit more complicated than size alone can handle.

In these cases, we can use the `replicate()` function to repeat an experiment many times.

The call `replicate(n, expr)` repeats the expression `expr` a total of `n` times and stores the results in a vector. Longer expressions are typically typed within curly braces `{}` so they can be read over several lines in R.

Constructing the Experiment

As we saw with `sample()`, it is helpful to construct an experiment in parts.

- ▶ Write code to conduct the experiment a single time.
- ▶ Replicate the code a small number of times (about 5)
- ▶ Replicate the code a large number of times (about 10000)
- ▶ Take the mean results of the event to calculate its probability.

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Multiple Dice Rolls, Pt 1

Simulate rolling a dice 7 times and computing the sum of all rolls and recording if the sum is greater than 30.

```
dice <- sample(___, size=___,replace=___)  
___(____)>_____
```

First we want to define a single experiment before moving to replicate it.

Multiple Dice Rolls, Pt 2

Let's run this experiment 5 times.

```
replicate(n=___, {  
  dice <- sample(___, size=___,replace=___)  
  ___(____)>____  
})
```

Multiple Dice Rolls, Pt 3

We will almost always want to replicate things a large number of times, say $n = 10000$. We then store the output in a vector.

```
results_dice <- replicate(n=___, {  
  dice <- sample(___, size=___,replace=___)  
  ___(____)>____  
})
```

Multiple Dice Rolls, Pt 4

Now, calculate the probability of rolling a die 7 times and getting a sum larger than 30.

```
results_dice <- replicate(n=___, {  
  dice <- sample(___, size=___, replace=___)  
  ___(____)>____  
})  
mean(_____)
```

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Exercise 1: Dice, pt. 1

If two die are rolled, what is the probability that the difference between the two numbers is less than 3?

First consider what the theoretical value would be (how many outcomes out of how many result in such a difference).

Exercise 1: Dice, pt. 2

If two dice are rolled, what is the probability that the difference between the two numbers is less than 3?

Now let's code a way to simulate this problem. Remember: one trial first, then replicate a few times, then replicate many times.

Hint: There is a function called `abs()` that can take the absolute value—useful for finding a difference less than 3 between two die rolls regardless of order.

Exercise 2: Coin, pt. 1

A fair coin is repeatedly tossed ten times. Compute the probability that the last three coin tosses results in heads. (Hint, review Example 2.13 in the textbook for an example)

Once again, let's first find the theoretical value.

Exercise 2: Coin, pt. 2

A fair coin is repeatedly tossed ten times. Compute the probability that the last three coin tosses results in heads.

Now let's find the value through simulation. You may want to sample from a vector consisting of just "H" and "T" to begin this problem.

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Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.